

CS 31

Discrete Mathematics for Computer Science II

COMPLETE MASTER STUDY GUIDE

Topic: Discrete Probability

Definitions · Theorems · Concepts · Exam Walkthroughs

Covers: Long Exam 2 | AY 2025–2026, 2nd Semester

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Foundations: Sample Spaces & Probability

The language and axioms of probability

1.1 Core Definitions

Definition 1 — Experiment, Sample Space, Event

An **experiment** is any procedure that yields one of a set of possible outcomes.

The **sample space** S is the complete set of possible outcomes.

An **event** E is any subset of S ($E \subseteq S$).

Definition 2 — Laplace's Definition (Equally Likely Outcomes)

When S is finite and all outcomes are equally likely:

$$p(E) = |E| / |S|$$

Since $E \subseteq S$, we always have $0 \leq p(E) \leq 1$.

Definition 3 — Probability Distribution (Unequal Outcomes)

Assign probability $p(s)$ to each outcome $s \in S$ such that:

- (i) $0 \leq p(s) \leq 1$ for every $s \in S$
- (ii) $\sum p(s) = 1$ (sum over all $s \in S$)

The probability of event E is then: $p(E) = \sum p(s)$ for $s \in E$

Definition 4 — Uniform Distribution

If $|S| = n$ and all outcomes are equally likely, each gets probability $1/n$. This is the **uniform distribution**.

1.2 Probability Theorems for Derived Events

Theorem 1 — Complement Rule

For any event E with complement $\bar{E} = S - E$:

$$p(\bar{E}) = 1 - p(E)$$

Use when it is easier to count outcomes NOT in E .

Theorem 2 — Inclusion-Exclusion (Union)

For any two events E_1 and E_2 :

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

If E_1 and E_2 are **disjoint** (mutually exclusive), $p(E_1 \cap E_2) = 0$, so just add.

Theorem 3 — Union of Pairwise Disjoint Events

If $E_1, E_2, \dots$ are pairwise disjoint (no two share an outcome):

$$p(E_1 \cup E_2 \cup \dots) = p(E_1) + p(E_2) + \dots$$

Exam Q1 Preview — Biased Coin (Solved on Exam)

Problem: Special coin: $P(H \text{ on 2nd flip} \mid 1st \text{ was } T)$ is 0.6; otherwise 0.5. Find $P(H_2)$.

Define H_1 =1st flip head, T_1 =1st flip tail, H_2 =2nd flip head.

Use Law of Total Probability:

$$\begin{aligned} P(H_2) &= P(H_2 \mid T_1) \cdot P(T_1) + P(H_2 \mid H_1) \cdot P(H_1) \\ &= 0.6 \times 0.5 + 0.5 \times 0.5 = 0.30 + 0.25 = \mathbf{0.55} \checkmark \text{ (Score: 10/10)} \end{aligned}$$

■ KEY INSIGHT / EXAM TIP

When a problem asks for an 'overall' or 'unconditional' probability, always check whether the Law of Total Probability applies: split on all mutually exclusive, exhaustive conditioning events.

Conditional Probability & Independence

How new information changes probabilities

Definition 6 — Conditional Probability

For events E and F with $p(F) > 0$, the probability of E *given* F is:

$$p(E | F) = p(E \cap F) / p(F)$$

Rearranged as the **Multiplication Rule**: $p(E \cap F) = p(E|F) \cdot p(F)$

Law of Total Probability

Since $E = (E \cap F) \cup (E \cap F^c)$ and these are disjoint:

$$p(E) = p(E|F) \cdot p(F) + p(E|F^c) \cdot p(F^c)$$

Generalized: if $F_1, \dots, F_n$ partition S, then $p(E) = \sum p(E|F_i) \cdot p(F_i)$

Definition 7 — Independent Events

Events E and F are **independent** if and only if:

$$p(E \cap F) = p(E) \cdot p(F)$$

Equivalently: $p(E|F) = p(E)$ and $p(F|E) = p(F)$.

Definition 8 — Pairwise vs Mutual Independence

Pairwise independent: $p(E_i \cap E_j) = p(E_i) \cdot p(E_j)$ for every pair $i \neq j$.

Mutually independent: $p(E_1 \cap \dots \cap E_n) = p(E_1) \cdot \dots \cdot p(E_n)$.

Mutual independence is stronger. Mutually independent $\implies$ pairwise independent, but NOT vice versa.

■ COMMON MISTAKE

'Pairwise independent' does NOT imply 'mutually independent'. There exist sets of events that are pairwise independent but not mutually independent.

Exam Q4 — RRPS Card Game (Score: 10/10)

Setup: You have 4 Rocks, 1 Paper, 1 Scissors. Rival has 2 Rocks, 3 Papers, 1 Scissors. Find $P(\text{you win a single match})$.

Winning combos (E): $\{\text{Rock} \cap \text{Scissors}, \text{Paper} \cap \text{Rock}, \text{Scissors} \cap \text{Paper}\}$

By disjoint union + independence of card selections:

$$P(E) = P(R_1)P(S_2) + P(P_1)P(R_2) + P(S_1)P(P_2)$$

$$= (4/6)(1/6) + (1/6)(2/6) + (1/6)(3/6)$$

$$= 4/36 + 2/36 + 3/36 = 9/36 = \mathbf{1/4} \checkmark$$

■ KEY INSIGHT / EXAM TIP

Card selections are independent because both players reveal simultaneously and randomly. Always verify independence before multiplying probabilities.

Chapter 3

Bayes' Theorem

Updating beliefs with new evidence

Theorem 4 — Bayes' Theorem

For events E and F with $p(E) \neq 0$ and $p(F) \neq 0$:

$$p(F|E) = [p(E|F) \cdot p(F)] / [p(E|F) \cdot p(F) + p(E|F^c) \cdot p(F^c)]$$

The denominator is just $p(E)$ by the Law of Total Probability.

Theorem 5 — Generalized Bayes' Theorem

If $F_1, \dots, F_n$ are mutually exclusive with $\cup F_i = S$:

$$p(F_j|E) = p(E|F_j) \cdot p(F_j) / \sum_i p(E|F_i) \cdot p(F_i)$$

How to Apply Bayes' Theorem — Step by Step

Step	Action	Formula / Note
1	Identify what you want to find	$p(\text{cause} \text{observed effect})$
2	Name the events clearly	$E = \text{observed evidence}, F = \text{hypothesis}$
3	List all given probabilities	$p(F), p(F^c), p(E F), p(E F^c)$
4	Compute denominator = $p(E)$	$p(E F) \cdot p(F) + p(E F^c) \cdot p(F^c)$
5	Divide numerator by $p(E)$	$p(E F) \cdot p(F) / p(E)$

Exam Q2 — Easter Salubong / Sister's Nap (Score: 10/10)

Given: Sister naps ≥ 3 hrs ($N_{\geq 3}$, $p=0.1$), 1–3 hrs ($N_{1,3}$, $p=0.7$), ≤ 1 hr ($N_{\leq 1}$, $p=0.2$).

$P(\text{awake} | N_{\geq 3}) = 0.8$, $P(\text{awake} | N_{1,3}) = 0.6$, $P(\text{awake} | N_{\leq 1}) = 0.2$

Find: $P(N_{\geq 3} | \text{not awake})$ — she missed the procession; find $P(\text{napped} \geq 3 \text{ hrs})$.

$$P(\text{not awake} | N_{\geq 3}) = 0.2, P(\text{not awake} | N_{1,3}) = 0.4, P(\text{not awake} | N_{\leq 1}) = 0.8$$

$$\text{Denominator} = 0.2(0.1) + 0.4(0.7) + 0.8(0.2) = 0.02 + 0.28 + 0.16 = 0.46$$

$$P(N_{\geq 3} | \text{not awake}) = 0.2 \times 0.1 / 0.46 = 0.02 / 0.46 \approx \mathbf{0.0435} \checkmark$$

■ KEY INSIGHT / EXAM TIP

The 'counter-intuitive' result makes sense: if she slept only a little (≤ 1 hr), she was most likely to miss the event. So observing 'she missed it' actually makes the long-nap scenario less likely.

Random Variables & Expected Value

Assigning numbers to outcomes; the long-run average

Definition 9 — Random Variable

A **random variable** X is a function $X: S \rightarrow \mathbb{R}$ that assigns a real number to each outcome in the sample space. If the range is a subset of integers, X is a **discrete random variable**.

Definition 10 & 11 — Distribution and Expected Value

The **distribution** of X is the set of pairs $\{(r, p(X=r)) : r \in \text{range}(X)\}$.

The **expected value** (mean) of X is:

$$E(X) = \sum_{s \in S} p(s) \cdot X(s) = \sum_r r \cdot p(X=r)$$

Theorem 6 — Expected Value via Distribution

$$E(X) = \sum_{r \in \text{range}(X)} r \cdot p(X=r)$$

This groups outcomes by their X -value, which is often faster than summing over all s .

Theorem 7 — Linearity of Expectation

$$(i) E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$(ii) E(aX + b) = a \cdot E(X) + b$$

Linearity holds even when X_i are not independent. This is extremely powerful!

Theorem 8 — Product of Independent Variables

If X and Y are **independent** random variables:

$$E(XY) = E(X) \cdot E(Y)$$

This does NOT hold in general — independence is required.

Exam Q3 — Consecutive Card Pairs (Score: 1/10 — See Mistake Analysis)

Problem: Standard deck of 52. Find $E(X)$ = expected number of pairs of consecutive cards with the same face value.

Correct Approach using Indicator Variables:

Let $X_i = 1$ if cards i and $i+1$ have the same face value ($i=1\ldots 51$). Then $X = \sum X_i$.

$P(X_1=1) = P(\text{card 2 matches card 1}) = 3/51$ (3 of remaining 51 cards match).

By linearity: $E(X) = 51 \times (3/51) = 3$

Why the exam answer was marked low: The student's formula became extremely complex. The clean indicator variable approach gives the answer in two lines.

■ KEY INSIGHT / EXAM TIP

Indicator Variable Trick: For 'how many X satisfy condition C ', define $X_i = 1$ if condition holds for item i , else 0. Then $E(\sum X_i) = \sum E(X_i) = \sum P(\text{condition holds for } i)$. This bypasses complex combinatorics.

Variance & Standard Deviation

How spread out is the distribution?

Definition 13 & 14 — Deviation and Variance

The **deviation** of X at outcome s is $X(s) - E(X)$.

The **variance** of X is the expected squared deviation:

$$V(X) = \sum_{s \in S} (X(s) - E(X))^2 \cdot p(s)$$

The **standard deviation** is $\sigma(X) = \sqrt{V(X)}$.

Theorem 9 — Computational Formula for Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Always use this formula in calculations — squaring then subtracting is much faster than computing each squared deviation.

Theorem 10 — Bienaymé's Formula

For **pairwise independent** random variables $X_1, \dots, X_n$:

$$V(X_1 + \dots + X_n) = V(X_1) + \dots + V(X_n)$$

Also: $V(aX + b) = a^2 \cdot V(X)$ and $V(X - Y) = V(X) + V(Y)$ (when independent).

5.1 Important Variance Note on Independence

Bienaymé's Formula requires **pairwise independence**. Linearity of Expectation (Thm 7) has NO such requirement. This distinction is the source of many exam errors.

- $E(X+Y) = E(X)+E(Y) \rightarrow$ always true
- $V(X+Y) = V(X)+V(Y) \rightarrow$ only true when X, Y are independent

Chapter 6

Bernoulli Trials & Binomial Distribution

Repeated independent success/failure experiments

Definition 15 — Bernoulli Trial

An experiment with exactly two outcomes (**success** with probability p , **failure** with probability $q = 1-p$). Each performance is one Bernoulli trial.

Trials are **mutually independent**: the outcome of any trial does not affect others.

Theorem 11 — Binomial Probability

The probability of exactly k successes in n independent Bernoulli trials:

$$P(X = k) = C(n, k) \cdot p^k \cdot (1-p)^{n-k}$$

where $C(n, k) = n! / (k!(n-k)!)$ is the binomial coefficient.

Theorem 12 — Expected Successes in Bernoulli Trials

$$E(X) = np$$

Proof via indicator variables: let $X_i = 1$ if trial i succeeds. $E(X_i) = p$, so $E(X) = np$.

$$\text{Variance of Binomial: } V(X) = np(1-p) = npq$$

For a single Bernoulli trial: $E(X) = p$, $V(X) = p(1-p) = pq$.

Exam Q7 — At Most 5 Failures Before 10 Successes (Score: 8/10)

Problem: $P(\text{at most 5 failures before achieving 10 successes}) = ?$

Model: Negative Binomial. Total trials $n = \text{failures} + 10 \text{ successes} = f + 10$.

'At most 5 failures' means the 10th success occurs on trial 10, 11, 12, 13, 14, or 15.

Equivalently: in $n = 10+k$ trials, there are exactly 10 successes and k failures ($k=0..5$), and the last trial must be a success:

$$P(X=k \text{ failures}) = C(10+k-1, k) \cdot p^{10} \cdot q^k = C(9+k, k) \cdot p^{10} \cdot q^k$$

$$\text{Total: } P = \sum_{k=0}^5 C(9+k, k) \cdot p^{10} \cdot q^k = \sum_{k=0}^5 C(10+k, 10) \cdot p^{10} \cdot q^k$$

The student's approach using the Binomial directly was partially correct but overcounted. The clean form shown above avoids double-counting.

■ KEY INSIGHT / EXAM TIP

Negative Binomial: $P(\text{r-th success on trial } n) = C(n-1, r-1) \cdot p^r \cdot q^{n-r}$. 'At most f failures before r successes' sums this over $n = r$ to $r+f$.

Chapter 7

Discrete Probability Distributions

Uniform, Bernoulli, Binomial, Geometric, Poisson

Distribution	PMF $p(X=r)$	$E(X)$	$V(X)$	When to Use
Uniform(n)	$1/n$ for $r=1..n$	$(n+1)/2$ (if $1..n$)	$(n^2-1)/12$	Each outcome equally likely (dice, random selection)
Bernoulli(p)	p if $r=1$; $1-p$ if $r=0$	p	$p(1-p)=pq$	Single success/failure trial
Binomial(n,p)	$C(n,k) \cdot p^k \cdot q^{n-k}$	np	npq	Fixed n independent trials; count successes
Geometric(p)	$(1-p)^{r-1} \cdot p$ $r \geq 1$	$1/p$	$(1-p)/p^2$	Count trials until 1st success
Poisson(λ)	$\lambda^x \cdot e^{-\lambda} / x!$ $x \geq 0$	λ	λ	Count rare events; approx Bin(n,p) when n large, p small, $\lambda=np$

Definition 19 — Geometric Distribution

X = number of independent Bernoulli trials needed to get the **first success**.

$$\text{PMF: } p(X = r) = (1-p)^{r-1} \cdot p \text{ for } r = 1, 2, 3, \dots$$

$$E(X) = 1/p \quad V(X) = (1-p)/p^2$$

Definition 20 — Poisson Distribution

$X \sim \text{Poi}(\lambda)$ when:

$$p(X = x) = \lambda^x \cdot e^{-\lambda} / x! \text{ for } x = 0, 1, 2, \dots$$

$E(X) = \lambda$, $V(X) = \lambda$. Approximates Bin(n,p) when $n \geq 30$, $p \leq 0.05$, $\lambda=np$.

Exam Q5 — Dice Betting Game (Score: 9.5/10)

Setup: Roll 3 dice. Bet on a number. Cost=80. Win 160 if appears once, 180 twice, 320 all three.

Net gains: A=80(+1), B=100(+2), C=240(+3), D=-80(none). $p=1/6$, $q=5/6$.

$$P(A) = C(3,1)(1/6)(5/6)^2 = 3 \cdot (1/6) \cdot (25/36) = 75/216$$

$$P(B) = C(3,2)(1/6)^2(5/6) = 3 \cdot (1/36) \cdot (5/6) = 15/216$$

$$P(C) = (1/6)^3 = 1/216$$

$$P(D) = (5/6)^3 = 125/216$$

$$E(X) = 80(75/216) + 100(15/216) + 240(1/216) - 80(125/216)$$

$$= (6000+1500+240-10000)/216 = -2260/216 \approx -10.46 \text{ (expected loss per game)}$$

Deduction: Answer marked 9.5 because net gain values need to be consistent with the problem (cost already deducted).

Covariance

Measuring linear dependence between two random variables

Covariance — Definition

For random variables X and Y:

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

Key Covariance Identities (Part (a) of Exam Q8)

$$(a) \text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

Proof:

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E(X))(Y - E(Y))] \\ &= E[XY - X \cdot E(Y) - Y \cdot E(X) + E(X) \cdot E(Y)] \\ &= E(XY) - E(Y) \cdot E(X) - E(X) \cdot E(Y) + E(X) \cdot E(Y) \text{ (by linearity)} \\ &= E(XY) - E(X) \cdot E(Y) \blacksquare\end{aligned}$$

Part (b): Independence Implies Zero Covariance

If X and Y are **pairwise independent**, then by Theorem 8: $E(XY) = E(X) \cdot E(Y)$.

$$\text{Therefore: } \text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y) = 0$$

Warning: $\text{Cov}(X, Y) = 0$ does NOT imply independence (zero covariance is necessary but not sufficient for independence).

Exam Q8 — Covariance Proof (Score: 1/10 — Analysis)

Grader's issue: The student's proof of part (a) circularly applied linearity of E to a product, incorrectly writing $E[X - E(X)] \cdot E[Y - E(Y)]$ (that would be E of a sum, not a product). The correct expansion expands the product inside E first, then applies linearity.

Correct proof structure:

1. Expand $(X - E(X))(Y - E(Y)) = XY - X \cdot E(Y) - Y \cdot E(X) + E(X) \cdot E(Y)$
2. Apply linearity to each term: $E(XY) - E(X)E(Y) - E(Y)E(X) + E(X)E(Y)$
3. Simplify: $E(XY) - E(X)E(Y) \blacksquare$

■ **COMMON MISTAKE**

$E[(X-\mu_X)(Y-\mu_Y)] \neq E[X-\mu_X] \cdot E[Y-\mu_Y]$. The product is inside ONE expectation. You must expand the product algebraically before applying linearity.

Chapter 9

Chebyshev's Inequality & Law of Large Numbers

Bounding deviations; convergence of sample averages

Theorem 18 — Chebyshev's Inequality

For any random variable X and positive real r :

$$P(|X - E(X)| \geq r) \leq V(X) / r^2$$

Gives an upper bound on how likely X is to deviate from its mean by at least r .

Theorem 19 — Weak Law of Large Numbers

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 . Let $S_n = \sum X_i$. For any $\epsilon > 0$:

$$P(|S_n/n - \mu| \geq \epsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

The sample average converges in probability to the true expected value.

Intuition: In a casino, you might win or lose on any single play. But if you play thousands of times, your average gain per game gets very close to $E(X)$ — which for most casino games is negative. This is why 'the house always wins.'

Exam Q6 — Store Price Hike (Score: 3/10 — Analysis)

Problem: 60% essentials, 40% non-essentials. 25% of essentials have price hike, 55% of non-essentials have price hike. Given at least one of two randomly selected items had a price hike, find $P(\text{both are essentials})$.

Let E =essential, H =price hike. $P(H|E)=0.25$, $P(H|\neg E)=0.55$.

$$P(H \cap E) = 0.6 \times 0.25 = 0.15, P(H \cap \neg E) = 0.4 \times 0.55 = 0.22$$

For 2 items: Let H_a = at least one price hike, H_b = both price hiked.

$P(E_b | H_a)$ requires careful setup. The student's attempt used Bayes correctly in structure but with arithmetic errors in the denominator.

Key formula: $P(\text{both essential} \cap \text{at least one hike}) / P(\text{at least one hike}) = P(E_b | H_a)$

$P(\text{at least one hike}) = 1 - P(\text{neither hiked}) = 1 - [(1-0.15)(1-0.22)] \dots$ must enumerate all cases.

Complete Exam Walkthroughs

Every question from Long Exam 2 — fully explained

Score Summary

Q #	Topic	Score	Concept Tested	Key Technique
1	Biased Coin	10/10	Conditional prob + Law of Total Prob	Split on conditioning event
2	Easter Salubong (Bayes)	10/10	Bayes' Theorem (3 hypotheses)	Generalized Bayes
3	Consecutive Card Pairs	1/10	Expected value + Indicator variables	$X = \sum X_i$, $E(X_i) = P(\text{match})$
4	RRPS Card Game	10/10	Disjoint union + independence	List winning combos, multiply
5	Dice Betting	9.5/10	Binomial PMF + $E(X)$ computation	Sum over all outcome cases
6	Store Price Hike	3/10	Conditional prob with 2 items	Enumerate joint events
7	Failures Before Successes	8/10	Negative Binomial distribution	Fix 10th success on last trial
8	Covariance Proof	1/10	Algebraic proof of Cov formula	Expand product THEN apply E
9	Birthday Problem	3/10	Complement + inequality for $n=23$	$P(\text{no shared b-day})$ product form
10	Runs in Bernoulli Trials	1/10	$E(R)$ via indicator variables	$E(I_j)$ for $j \geq 1 = p(1-p)$

Q9 — Birthday Problem (Score: 3/10)

Birthday Problem — Full Correct Solution

Goal: Find minimum n such that $P(\geq 2 \text{ people share a birthday}) \geq 0.5$.

It is easier to find $P(\text{no shared birthday}) = P(\text{all } n \text{ birthdays distinct})$:

$$P(\text{no match}) = (365/365) \cdot (364/365) \cdot (363/365) \cdot \dots \cdot ((365-n+1)/365)$$

$$= 365 \cdot 364 \cdot 363 \cdot \dots \cdot (365-n+1) / 365^n$$

$$= P(365, n) / 365^n \text{ where } P(365, n) \text{ is the falling factorial.}$$

$$\text{We need: } 1 - P(\text{no match}) \geq 0.5 \quad \blacksquare \quad P(\text{no match}) \leq 0.5$$

$$\text{Numerically: } n = 23 \text{ gives } P(\text{no match}) \approx 0.4927, \text{ so } P(\text{match}) \approx 0.5073 \geq 0.5. \checkmark$$

Exam error: Student wrote $P(n=23) = (1-p)^{22} \cdot p$ using geometric distribution, which is WRONG. The birthday problem is NOT a geometric trial — each person adds a new constraint, not a repeat of the same trial.

Q10 — Runs in Bernoulli Trials (Score: 1/10)

Runs — Full Correct Solution

Definition: A 'run' = maximal consecutive sequence of successes. R = total number of runs in n trials.

Hint decomposition: $R = \sum I_j$ where $I_j = 1$ if a run begins at trial j .

A run begins at trial j if and only if trial j is a success AND ($j=1$ OR trial $j-1$ is a failure).

$$\text{For } j = 1: P(I_1=1) = p$$

$$\text{For } j > 1: P(I_j=1) = P(S \text{ at } j) \cdot P(F \text{ at } j-1) = p \cdot (1-p) = p \cdot q$$

$$\text{By linearity: } E(R) = E(I_1) + \sum_{j=2}^n E(I_j)$$

$$= p + (n-1) \cdot p \cdot q = p + (n-1)p(1-p)$$

Exam error: Student set $E(I_1) = 1 \cdot p + 0 \cdot (1-p) = 1$ (correct), but then set $E(I_j)$ for $j > 1$ ALSO = p , ignoring the additional condition that the previous trial must be a FAILURE. Also, the final sum was incorrectly simplified to n instead of the formula above.

Quick-Reference Formula Sheet

Every formula in one place — suitable for review

Probability Foundations

Rule	Formula	Notes
Laplace	$p(E) = E / S $	Equally likely outcomes only
Complement	$p(\blacksquare) = 1 - p(E)$	
Inclusion-Excl.	$p(E \cup F) = p(E) + p(F) - p(E \cap F)$	
Disjoint union	$p(E \cup F) = p(E) + p(F)$	When $E \cap F = \emptyset$
Conditional	$p(E F) = p(E \cap F)/p(F)$	$p(F) > 0$
Mult. Rule	$p(E \cap F) = p(E F) \cdot p(F)$	Rearranged conditional
Independence	$p(E \cap F) = p(E) \cdot p(F)$	Iff independent
Total Prob.	$p(E) = \sum p(E F_{\blacksquare}) \cdot p(F_{\blacksquare})$	$F_{\blacksquare} \dots F_{\blacksquare}$ partition S
Bayes	$p(F E) = p(E F)p(F) / [p(E F)p(F) + p(E F_{\blacksquare})p(F_{\blacksquare})]$	

Expected Value & Variance

Quantity	Formula	Conditions
$E(X)$	$\sum r \cdot p(X=r)$	Always
Linearity	$E(X_{\blacksquare} + \dots + X_{\blacksquare}) = E(X_{\blacksquare}) + \dots + E(X_{\blacksquare})$	Always (no independence needed)
$E(aX+b)$	$a \cdot E(X) + b$	Always
$E(XY)$	$E(X) \cdot E(Y)$	X, Y independent only
$V(X)$	$E(X^2) - [E(X)]^2$	Computational formula
$V(aX+b)$	$a^2 \cdot V(X)$	
$V(X+Y)$	$V(X) + V(Y)$	X, Y independent (Bienaymé)
$\text{Cov}(X, Y)$	$E(XY) - E(X)E(Y)$	=0 when independent
Chebyshev	$P(X - E(X) \geq r) \leq V(X)/r^2$	Any $r > 0$

Distributions — PMF, Mean, Variance

Distribution	PMF	Mean	Variance
Bernoulli(p)	$p(1)=p, p(0)=1-p$	p	pq
Binomial(n, p)	$C(n, k)p^k q^{n-k}$	np	npq
Geometric(p)	$(1-p)^{r-1} \cdot p, r \geq 1$	$1/p$	q/p^2
Poisson(λ)	$\frac{\lambda^x e^{-\lambda}}{x!}, x \geq 0$	λ	λ
Uniform(n)	$1/n$ for $r=1..n$	$(n+1)/2$	$(n^2-1)/12$

Key Techniques for Exam Problems

Technique	When to Use	How to Apply
Indicator Variables	Count expected number of X that satisfy property	$X_i = 1$ if property holds. $E(X) = \sum P(\text{property holds for } i)$
Law of Total Probability	Unconditional prob when conditions split easily	$p(E) = \sum p(E F_i)p(F_i)$ over all F_i
Complement Rule	$P(\text{at least one}), P(\text{not all}), P(\text{some match})$	$p(\text{event}) = 1 - p(\text{complement})$
Bayes' Theorem	Given effect, find cause	Posterior = likelihood $\times$ prior / evidence
Linearity of Expectation	E of sum of random variables	Always split $X = \sum X_i$ and use $E(X) = \sum E(X_i)$
Binomial Model	n independent identical trials, count successes	$X \sim \text{Bin}(n, p), P(X=k) = C(n, k)p^k q^{n-k}$
Geometric Model	Count trials to first success	$P(X=r) = q^{r-1} \cdot p, E(X) = 1/p$
Disjoint Union	Mutually exclusive events	$P(A \cup B) = P(A) + P(B)$

Critical Distinctions (Exam Traps)

Common Error	Correct Understanding
Writing $E[X-\mu] \cdot E[Y-\mu]$ for $\text{Cov}(X, Y)$	Expand the product FIRST inside E , then apply linearity
Assuming pairwise indep. implies mutual indep.	Mutual $\implies$ Pairwise, but Pairwise $\not\implies$ Mutual
$V(X+Y) = V(X) + V(Y)$ always	Only when X, Y are (pairwise) independent
$E(XY) = E(X)E(Y)$ always	Only when X, Y are independent
Birthday problem = geometric distribution	Each new person adds a different conditional constraint

$E(I_j)$ for runs same for all j	$j=1: E=p; j>1: E=p(1-p)$ since prev. must be failure
Cov=0 implies independence	Zero covariance is necessary but not sufficient

Master Strategy for Discrete Probability Exams

How to approach any problem from scratch

Step 1 — READ CAREFULLY and define every random variable and event explicitly with notation. Points are lost for undefined variables.

Step 2 — CLASSIFY the problem type:

- 'Given ... what is the probability of ...' → Conditional probability or Bayes
- 'How many on average' / 'expected number' → $E(X)$ via indicator variables or distribution formula
- 'Fixed number of trials, count successes' → Binomial
- 'Count until first success' → Geometric
- 'Sum of random variables' → Linearity of Expectation

Step 3 — WRITE the formula before substituting numbers. Examiners award partial credit for correct formulas even with arithmetic errors.

Step 4 — CHECK: Does your answer make sense? Probabilities must be in $[0,1]$. Expected counts must be ≥ 0 . Variances must be ≥ 0 .

Step 5 — JUSTIFY every step. 'No justification, no points' is the explicit exam rule. State the theorem or definition you are applying.

■ KEY INSIGHT / EXAM TIP

The single most powerful technique on this exam is the **Indicator Variable / Linearity of Expectation** method. Whenever a problem asks for $E(X)$ and X counts something, immediately decompose $X = X_1 + X_2 + \dots + X_n$ where each X_i is a 0/1 indicator. Then $E(X) = \sum P(X_i = 1)$. This bypasses nearly all combinatorial complexity.

This technique solved Q1 (biased coin, used Law of Total Probability — same idea), Q3 (card pairs), Q4 (RRPS), and Q10 (runs) on this exact exam.